

Skill 1 DBSE & DBD

The screenshot shows the SQL Enterprise Manager interface. The left pane displays the 'Schemas' tree with 'bookflow_skill1' selected. The central pane shows the 'Query 1' window with the following SQL script:

```
1 CREATE DATABASE bookflow_Skill1;
2 USE bookflow_Skill1;
```

The right pane shows the 'SQLAdditions' window with a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help." The bottom pane shows the 'Output' window with the following results:

#	Time	Action	Message	Duration / Fetch
1	13:40:50	CREATE DATABASE bookflow_Skill1	1 row(s) affected	0.047 sec
2	13:40:53	USE bookflow_Skill1	0 row(s) affected	0.000 sec

1. Create Table

The screenshot shows the SQL Enterprise Manager interface. The left pane displays the 'Schemas' tree with 'bookflow_skill1' selected. The central pane shows the 'Query 1' window with the following SQL script:

```
1 CREATE DATABASE bookflow_Skill1;
2 USE bookflow_Skill1;
3
4 CREATE TABLE student_marks (
5     roll_no INT PRIMARY KEY,
6     name VARCHAR(50),
7     subject VARCHAR(50),
8     marks DECIMAL(5,2)
9 );
10
```

The right pane shows the 'SQLAdditions' window with a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help." The bottom pane shows the 'Output' window with the following results:

#	Time	Action	Message	Duration / Fetch
1	13:40:50	CREATE DATABASE bookflow_Skill1	1 row(s) affected	0.047 sec
2	13:40:53	USE bookflow_Skill1	0 row(s) affected	0.000 sec
3	13:41:39	CREATE TABLE student_marks (roll_no INT PRIMARY KEY, name VARCHAR...	0 row(s) affected	0.063 sec

2. Insert Data

The screenshot shows the SQL Enterprise Manager interface. The left pane displays the 'Schemas' tree with 'bookflow_skill1' selected. The central pane shows the 'Query 1' window with the following SQL script:

```
10
11 INSERT INTO student_marks (roll_no, name, subject, marks) VALUES
12 (1, 'Ravi', 'Math', 85.50),
13 (2, 'Sita', 'Math', 92.75),
14 (3, 'Anil', 'Math', 78.40),
15 (4, 'Priya', 'Math', 88.90),
16 (5, 'Vijay', 'Math', 80.25),
17 (6, 'subbusir', 'aws', 98.50),
18 (7, 'dwaraka', 'dbms', 95.75),
19 (8, 'ranjani', 'english', 97.40),
20 (9, 'kavithamam', 'aws1', 99.90),
21 (10, 'seetha', 'azure', 82.25);
22
23
```

The right pane shows the 'SQLAdditions' window with a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help." The bottom pane shows the 'Output' window with the following results:

#	Time	Action	Message	Duration / Fetch
1	13:40:50	CREATE DATABASE bookflow_Skill1	1 row(s) affected	0.047 sec
2	13:40:53	USE bookflow_Skill1	0 row(s) affected	0.000 sec
3	13:41:39	CREATE TABLE student_marks (roll_no INT PRIMARY KEY, name VARCHAR...	0 row(s) affected	0.063 sec
4	13:42:43	INSERT INTO student_marks (roll_no, name, subject, marks) VALUES (1, 'Ravi', Math...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.047 sec

3. Simple Select

The screenshot shows the SQL Developer interface. The 'Navigator' pane on the left displays the database schema, including 'bookflow_db' and 'bookflow_skill1'. The 'Query' pane shows a SQL query: `SELECT * FROM student_marks;`. The 'Result Grid' displays the following data:

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	78.40
4	Priya	Math	88.90
5	Vijay	Math	80.25

The 'Output' pane shows the execution log with the following actions:

#	Time	Action	Message	Duration / Fetch
1	13:40:50	CREATE DATABASE bookflow_skill1	1 row(s) affected	0.047 sec
2	13:40:53	USE bookflow_skill1	0 row(s) affected	0.000 sec
3	13:41:39	CREATE TABLE student_marks (roll_no INT PRIMARY KEY, name VARCHAR(100), subject VARCHAR(100), marks DECIMAL(5,2))	0 row(s) affected	0.063 sec
4	13:42:43	INSERT INTO student_marks (roll_no, name, subject, marks) VALUES (1, 'Ravi', 'Math', 85.50), (2, 'Sita', 'Math', 92.75), (3, 'Anil', 'Math', 78.40), (4, 'Priya', 'Math', 88.90), (5, 'Vijay', 'Math', 80.25)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.047 sec
5	13:45:28	SELECT * FROM student_marks LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

4. Aggregate Functions

The screenshot shows the SQL Developer interface. The 'Query' pane shows a SQL query: `SELECT COUNT(*) AS total_students FROM student_marks; SELECT SUM(marks) FROM student_marks; SELECT AVG(marks) FROM student_marks; SELECT MAX(marks) FROM student_marks; SELECT MIN(marks) FROM student_marks;`. The 'Result Grid' displays the following data:

MIN(marks)
78.40

The 'Output' pane shows the execution log with the following actions:

#	Time	Action	Message	Duration / Fetch
6	13:47:38	SELECT COUNT(*) AS total_students FROM student_marks LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
7	13:47:42	SELECT SUM(marks) FROM student_marks LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
8	13:47:45	SELECT AVG(marks) FROM student_marks LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
9	13:47:46	SELECT MAX(marks) FROM student_marks LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
10	13:47:47	SELECT MAX(marks) FROM student_marks LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
11	13:47:50	SELECT MIN(marks) FROM student_marks LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

5. Filtering with WHERE

The screenshot shows the SQL Developer interface. The 'Query' pane shows a SQL query: `SELECT * FROM student_marks WHERE marks < 80; SELECT * FROM student_marks WHERE marks BETWEEN 80 AND 90; SELECT * FROM student_marks WHERE name LIKE 'P%'; SELECT * FROM student_marks WHERE name IN ('Ravi', 'Sita', 'Vijay'); SELECT * FROM student_marks WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%');`. The 'Result Grid' displays the following data:

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
4	Priya	Math	88.90

The 'Output' pane shows the execution log with the following actions:

#	Time	Action	Message	Duration / Fetch
13	13:50:16	SELECT * FROM student_marks WHERE marks >= 90 LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
14	13:50:18	SELECT * FROM student_marks WHERE marks < 80 LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
15	13:50:20	SELECT * FROM student_marks WHERE marks BETWEEN 80 AND 90 LIMIT 0, 1...	4 row(s) returned	0.000 sec / 0.000 sec
16	13:50:21	SELECT * FROM student_marks WHERE name LIKE 'P%'; LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
17	13:50:23	SELECT * FROM student_marks WHERE name IN ('Ravi', 'Sita', 'Vijay') LIMIT 0, 10...	3 row(s) returned	0.000 sec / 0.000 sec
18	13:50:25	SELECT * FROM student_marks WHERE marks > 85 AND (subject = 'Math' OR na...	3 row(s) returned	0.000 sec / 0.000 sec

6. Update Queries

The screenshot shows the SQL Developer interface with a query editor containing several SQL statements. The 'Output' pane at the bottom displays the execution results of these queries.

```
32 SELECT * FROM student_marks WHERE marks >= 90;
33 SELECT * FROM student_marks WHERE marks < 80;
34 SELECT * FROM student_marks WHERE marks BETWEEN 80 AND 90;
35 SELECT * FROM student_marks WHERE name LIKE 'PK';
36 SELECT * FROM student_marks WHERE name IN ('Ravi', 'Sita', 'Vijay');
37 SELECT * FROM student_marks WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'PK');
38
39 UPDATE student_marks SET marks = 90.00 WHERE roll_no = 3;
40 UPDATE student_marks SET subject = 'Remedial Math' WHERE marks < 80;
```

Output:

#	Time	Action	Message	Duration / Fetch
1	13:58:43	USE bookflow_Skill1	0 row(s) affected	0.000 sec
2	13:58:51	UPDATE student_marks SET marks = 90.00 WHERE roll_no = 3	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec
3	13:58:54	UPDATE student_marks SET subject = 'Remedial Math' WHERE marks < 80	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0	0.000 sec

7. Delete Queries

The screenshot shows the SQL Developer interface with a query editor containing several SQL statements. The 'Output' pane at the bottom displays the execution results of these queries.

```
32 SELECT * FROM student_marks WHERE marks >= 90;
33 SELECT * FROM student_marks WHERE marks < 80;
34 SELECT * FROM student_marks WHERE marks BETWEEN 80 AND 90;
35 SELECT * FROM student_marks WHERE name LIKE 'PK';
36 SELECT * FROM student_marks WHERE name IN ('Ravi', 'Sita', 'Vijay');
37 SELECT * FROM student_marks WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'PK');
38
39 UPDATE student_marks SET marks = 90.00 WHERE roll_no = 3;
40 UPDATE student_marks SET subject = 'Remedial Math' WHERE marks < 80;
41
42 DELETE FROM student_marks WHERE roll_no = 5;
43 DELETE FROM student_marks WHERE marks < 75;
```

Output:

#	Time	Action	Message	Duration / Fetch
1	13:58:43	USE bookflow_Skill1	0 row(s) affected	0.000 sec
2	13:58:51	UPDATE student_marks SET marks = 90.00 WHERE roll_no = 3	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec
3	13:58:54	UPDATE student_marks SET subject = 'Remedial Math' WHERE marks < 80	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0	0.000 sec
4	13:59:46	DELETE FROM student_marks WHERE roll_no = 5	1 row(s) affected	0.047 sec
5	13:59:49	DELETE FROM student_marks WHERE marks < 75	0 row(s) affected	0.000 sec

8. Sorting

The screenshot shows the SQL Developer interface with a query editor containing several SQL statements. The 'Result Grid' pane displays the results of the queries, and the 'Output' pane shows the execution results.

```
44
45 SELECT * FROM student_marks ORDER BY marks ASC;
46 SELECT * FROM student_marks ORDER BY marks DESC;
47 SELECT * FROM student_marks ORDER BY name ASC;
48 SELECT * FROM student_marks ORDER BY name DESC;
```

Result Grid:

roll_no	name	subject	marks
6	subbusir	avis	98.50
2	Sita	Math	92.75
10	seetha	azure	82.25
1	Ravi	Math	85.50
8	ranjani	english	97.40

Output:

#	Time	Action	Message	Duration / Fetch
4	13:59:46	DELETE FROM student_marks WHERE roll_no = 5	1 row(s) affected	0.047 sec
5	13:59:49	DELETE FROM student_marks WHERE marks < 75	0 row(s) affected	0.000 sec
6	14:00:42	SELECT * FROM student_marks ORDER BY marks ASC LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
7	14:00:45	SELECT * FROM student_marks ORDER BY marks DESC LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
8	14:00:47	SELECT * FROM student_marks ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
9	14:00:49	SELECT * FROM student_marks ORDER BY name DESC LIMIT 0, 1000	9 row(s) returned	0.016 sec / 0.000 sec

9. Group By

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'SCHEMAS' tree with 'bookflow_skill1' selected. The central pane shows 'Query 1' with the following SQL code:

```
48 SELECT * FROM student_marks ORDER BY name DESC;
49
50 SELECT subject, SUM(marks) AS total_marks FROM student_marks GROUP BY subject;
51 SELECT subject, AVG(marks) AS avg_marks FROM student_marks GROUP BY subject;
52 SELECT subject, COUNT(*) AS num_students FROM student_marks GROUP BY subject;
53
```

The 'Result Grid' shows the results of the third query (line 52):

subject	num_students
Math	4
avis	1
cbms	1
english	1
avis1	1

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The bottom pane shows the 'Output' window with a table of execution results:

#	Time	Action	Message	Duration / Fetch
7	14:00:45	SELECT * FROM student_marks ORDER BY name DESC LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
8	14:00:47	SELECT * FROM student_marks ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
9	14:00:49	SELECT * FROM student_marks ORDER BY name DESC LIMIT 0, 1000	9 row(s) returned	0.016 sec / 0.000 sec
10	14:02:15	SELECT subject, SUM(marks) AS total_marks FROM student_marks GROUP BY subject	6 row(s) returned	0.000 sec / 0.000 sec
11	14:02:19	SELECT subject, AVG(marks) AS avg_marks FROM student_marks GROUP BY subject	6 row(s) returned	0.000 sec / 0.000 sec
12	14:02:22	SELECT subject, COUNT(*) AS num_students FROM student_marks GROUP BY subject	6 row(s) returned	0.000 sec / 0.000 sec

10. Having

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'SCHEMAS' tree with 'bookflow_skill1' selected. The central pane shows 'Query 1' with the following SQL code:

```
57 HAVING AVG(marks) > 90;
58
59 SELECT subject, COUNT(*) AS num_students
60 FROM student_marks
61 GROUP BY subject
62 HAVING COUNT(*) > 1;
```

The 'Result Grid' shows the results of the second query (line 59):

subject	num_students
Math	4

The right pane shows the 'SQLAdditions' tab with a message: 'Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.'

The bottom pane shows the 'Output' window with a table of execution results:

#	Time	Action	Message	Duration / Fetch
9	14:00:49	SELECT * FROM student_marks ORDER BY name DESC LIMIT 0, 1000	9 row(s) returned	0.016 sec / 0.000 sec
10	14:02:15	SELECT subject, SUM(marks) AS total_marks FROM student_marks GROUP BY subject	6 row(s) returned	0.000 sec / 0.000 sec
11	14:02:19	SELECT subject, AVG(marks) AS avg_marks FROM student_marks GROUP BY subject	6 row(s) returned	0.000 sec / 0.000 sec
12	14:02:22	SELECT subject, COUNT(*) AS num_students FROM student_marks GROUP BY subject	6 row(s) returned	0.000 sec / 0.000 sec
13	14:03:11	SELECT subject, AVG(marks) AS avg_marks FROM student_marks GROUP BY subject	4 row(s) returned	0.000 sec / 0.000 sec
14	14:03:17	SELECT subject, COUNT(*) AS num_students FROM student_marks GROUP BY subject	1 row(s) returned	0.000 sec / 0.000 sec